PCBC H Parametric & Sequences Test (Practice B) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A baseball is hit from a height of 4 feet at an angle of 30° with an initial speed of 96 ft/s.

a) Write a set of parametric equations for the motion of the baseball. (Recall: $x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the baseball.

c) Determine how long the baseball is in the air.

d) Determine how far the baseball traveled in the air.

Write the first five terms of the sequence. (3 points)

8. $\{a_n\} = \left\{ \dfrac{n+1}{n^2} \right\}$

A sequence is defined recursively. Write the first five terms. (3 points)

9. $a_1 = 2; \quad a_n = 3a_{n-1} - 1$

Write out the sum. (3 points)

10. $\displaystyle\sum_{k=1}^{n} \frac{2k-1}{k}$

Express the sum using summation notation. (3 points)

11. $\dfrac{2}{5} + \dfrac{3}{6} + \dfrac{4}{7} + \cdots + \dfrac{20}{23}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#12 is 3 points and #13 is 4 points)

12. $\sum_{k=1}^{20} \left(3k^2 + 2k\right)$

13. $\sum_{k=5}^{15} 2k^3$

Expand using the Binomial Theorem. (4 points)

14. $(3x+2)^4$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Baseball: $h = 4$ ft, $\theta = 30°$, $v = 96$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (96\cos 30°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (96\sin 30°)t - 16t^2 + 4$ — *+h: starts at 4 ft*

$x_1(t) = 96\cos(30°)\cdot t,\ \ y_1(t) = 96\sin(30°)\cdot t - 16t^2 + 4$ — *what you type in*

$0 \le t \le 4$, tstep 0.01; x : 0 to 300, y : 0 to 50 — *graph doesn't come back down? raise tmax*

b) $\boxed{y \approx 40.000 \text{ ft}}$ — *trace to the top ($t \approx 1.500$); not Analyze, Maximum*

c) $\boxed{t \approx 3.081 \text{ s}}$ — *trace to $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 256.161 \text{ ft}}$ — *same point, read x*

Other way for c): $0 = -16t^2 + (96\sin 30°)t + 4$, quadratic formula, keep the positive t. Units: feet.

8. $\left\{\frac{n+1}{n^2}\right\}$

$\frac{2}{1}, \frac{3}{4}, \frac{4}{9}, \frac{5}{16}, \frac{6}{25}$

$\boxed{2, \frac{3}{4}, \frac{4}{9}, \frac{5}{16}, \frac{6}{25}}$

plug in $n = 1$ to 5; list with commas, no ellipsis

9. $a_1 = 2;\ a_n = 3a_{n-1} - 1$

$a_2 = 3(2) - 1 = 5$ — *3 times the previous term, minus 1*

$a_3 = 3(5) - 1 = 14,\ \ a_4 = 41,\ a_5 = 122$

$\boxed{2,\ 5,\ 14,\ 41,\ 122}$

10. $\sum_{k=1}^{n} \frac{2k-1}{k} = \boxed{1 + \frac{3}{2} + \frac{5}{3} + \cdots + \frac{2n-1}{n}}$

$k = 1, 2, 3$, then the last term with $k = n$. Don't add.

11. $\boxed{\sum_{k=1}^{19} \frac{k+1}{k+4}}$

tops 2 to 20 are $k+1$; bottom is 3 more. $k = 1$ gives $\frac{2}{5}$; $k = 19$ gives $\frac{20}{23}$.

12.

$3\sum_{k=1}^{20} k^2 + 2\sum_{k=1}^{20} k$ — *(2), (1)*

$3\left[\frac{20(21)(41)}{6}\right] + 2\left[\frac{20(21)}{2}\right]$ — *(7), (6); then calculator*

$8610 + 420 = \boxed{9030}$

13.

$2\left[\sum_{k=1}^{15} k^3 - \sum_{k=1}^{4} k^3\right]$ — *(1), (4): lower index 5, so $j = 4$*

$2\left[\left(\frac{15(16)}{2}\right)^2 - \left(\frac{4(5)}{2}\right)^2\right]$ — *(8), whole bracket squared*

$2(14400 - 100) = \boxed{28600}$

14. $(3x+2)^4$: row 1 4 6 4 1

$1(3x)^4 + 4(3x)^3(2) + 6(3x)^2(2)^2 + 4(3x)(2)^3 + 1(2)^4$ — *blanks first: 3x powers down, 2 up*

$81x^4 + 4(27x^3)(2) + 6(9x^2)(4) + 4(3x)(8) + 16$ — $(3x)^4 = 81x^4$

$\boxed{81x^4 + 216x^3 + 216x^2 + 96x + 16}$ — *second term positive: no alternating*